

Erysipelothrix rhusiopathiae Infectious Disease — Comprehensive Disease Characterization

Target disease: Erysipelothrix rhusiopathiae infectious disease **MONDO ID:** MONDO:0006752 · **ICD-10:** A26 (Erysipeloid) · **MeSH:** D004887 (Erysipeloid) / D004886 (Erysipelothrix Infections) **Category:** Infectious disease (bacterial zoonosis)

Summary

Erysipelothrix rhusiopathiae infectious disease is an occupationally acquired bacterial zoonosis caused by the Gram-positive bacillus *Erysipelothrix rhusiopathiae*. It is not a genetic disease; there are no causal genes, pathogenic variants, or inheritance pattern. The "genetic/molecular" dimension of this entry concerns the *pathogen's* virulence genes and the *host's* predisposing comorbidities rather than germline human variation. Human disease presents in three well-established clinical forms whose severity spans a wide range: (1) a localized cutaneous infection known as **erysipeloid** (the most common form, typically a self-limited hand/finger lesion acquired by direct inoculation), (2) a **diffuse/generalized cutaneous** form, and (3) a rare but dangerous **septicemic** form frequently complicated by **infective endocarditis** with case-fatality exceeding 38%.

Pathogenesis is driven principally by two virulence factors: **surface protective antigen A (SpaA)**, an adhesin of the Spa family, and **neuraminidase**, an enzyme that mediates tissue spread — whereas the capsule and hyaluronidase are *not* required for virulence. Progression from a localized skin lesion to invasive, disseminated disease is strongly gated by host susceptibility, particularly immunocompromising conditions such as diabetes mellitus and alcoholic cirrhosis. The organism is intrinsically **resistant to vancomycin**, an important clinical pitfall, because empiric vancomycin for Gram-positive bacteremia will fail; **penicillin and ceftriaxone are the drugs of choice**, with excellent in-vitro activity.

E. rhusiopathiae is a broad-host-range pathogen/commensal of mammals, birds, and fish; **swine erysipelas** is the animal disease of greatest prevalence and economic importance and drives much of the veterinary vaccine effort. Control in humans rests on occupational hygiene (protective gloves, wound care among butchers, fishermen, veterinarians, farmers) and, in animals, on **Spa-family (SpaA/SpaC) subunit and attenuated vaccines**, which are well validated in mouse and pig challenge models. This report consolidates 9 confirmed findings across 23 reviewed papers into a knowledge-base-ready characterization spanning all 15 requested domains, flagging clearly where a domain is not applicable to a non-genetic infectious disease.

1. Disease Information

Erysipelothrix rhusiopathiae infectious disease is a zoonotic bacterial infection of humans acquired occupationally through direct contact with colonized/infected animals, their products or wastes, or contaminated soil and water (including marine environments, where the organism persists long-term).

Key identifiers:

Resource	Identifier
MONDO	MONDO:0006752
ICD-10	A26 (Erysipeloid); A26.7 (Erysipelothrix sepsis); A26.9 (unspecified)
ICD-11	1B72 (Erysipeloid)
MeSH	D004887 (Erysipeloid); D004886 (Erysipelothrix Infections)
NCBI Taxonomy (pathogen)	txid1648 (<i>Erysipelothrix rhusiopathiae</i>)
OMIM / Orphanet	Not applicable (non-genetic infectious disease)

Synonyms / alternative names: Erysipeloid; erysipeloid of Rosenbach; fish-handler's disease; fish poisoning/"fish hand"; seal finger (overlapping usage); whale finger; Erysipelothrix infection; (in animals) swine erysipelas, "diamond skin disease."

Source of information: This entry is derived from **aggregated disease-level resources** — authoritative narrative reviews, microbiological studies, veterinary vaccinology, and individual clinical case reports — rather than from a single EHR/patient-level dataset. Because invasive human disease is rare, much of the clinical literature is case reports and small series.

2. Etiology

Disease causal factor: The disease is **infectious**. The sole etiologic agent is *Erysipelothrix rhusiopathiae*, a small, Gram-positive, non-spore-forming, catalase-negative bacillus. There is no genetic (host germline) causation.

Risk factors (environmental/occupational): - **Occupational exposure** is the dominant risk factor. Two authoritative reviews establish that human infection is acquired via contact with animals, their products/wastes, or soil (PMID: 19733019; PMID: 10482289). High-risk occupations include butchers, abattoir workers, fishermen and fish handlers, veterinarians, farmers, and poultry/swine workers. - **Direct skin inoculation** through cuts/abrasions on the hands is the typical portal for erysipeloid (PMID: 26693283). - **Host immunocompromise** predisposes to invasive/systemic disease: documented comorbidities include diabetes mellitus, alcoholic cirrhosis, and chronic alcohol abuse (PMID: 24772603).

Genetic risk factors: Not applicable — no human susceptibility loci, modifier genes, or causal variants are described for this infection.

Protective factors: Behavioral/occupational — protective gloves, hand hygiene, prompt wound care, and (in the animal reservoir) vaccination reduce transmission. No genetic protective variants are relevant.

Gene–environment interactions: Not applicable in the human-genetic sense. The functional analogue is **pathogen-genotype × host-status interaction**: virulent bacterial genotypes (SpaA⁺, neuraminidase-high) disseminate preferentially in immunocompromised hosts.

3. Phenotypes

E. rhusiopathiae produces three clinical forms (F001), summarized below with suggested HPO terms.

Phenotype	Type	Onset / course	Frequency	HPO suggestion
Erysipeloid (localized violaceous, well-demarcated, painful/pruritic plaque, usually hand/finger; spreads peripherally with central clearing)	Clinical sign / skin manifestation	Acute, days after inoculation; usually self-limited over ~2–4 weeks	Most common form	HP:0000988 (Skin rash); HP:0011368 (Dermatological manifestations); HP:0100659 (Abnormal skin morphology)
Local pain / burning / pruritus at lesion	Symptom	Acute	Common in erysipeloid	HP:0025280 (Pain); HP:0000989 (Pruritus)
Tenosynovitis of a digit	Clinical sign (complication of erysipeloid)	Subacute extension	Rare complication	HP:0100279 (Tenosynovitis)
Diffuse / generalized cutaneous form (widespread violaceous lesions ± systemic symptoms)	Physical manifestation	Subacute	Uncommon	HP:0000989 (Pruritus); HP:0011368
Septicemia / bacteremia (fever, malaise)	Clinical sign / lab abnormality	Acute–subacute	Rare	HP:0002837 (Bacteremia); HP:0001945 (Fever)
Infective endocarditis (often native, previously normal valves; valve	Clinical sign / complication	Subacute; high morbidity	Rare but severe	HP:0100584 (Endocarditis); HP:0001635 (Congestive heart failure, from

Phenotype	Type	Onset / course	Frequency	HPO suggestion
destruction; embolic events — stroke, retinal artery occlusion)				valve destruction); HP:0001297 (Stroke)
Musculoskeletal (psoas abscess, vertebral osteomyelitis, discitis, pyogenic spondylitis)	Clinical sign / complication	Subacute– chronic	Rare, in immunocompromised	HP:0002754 (Osteomyelitis); HP:0003418 (Back pain)

Age of onset: Adult-onset, reflecting occupational exposure; no congenital form. **Severity:** highly variable — from mild self-limited erysipeloid to lethal endocarditis. **Progression:** erysipeloid is typically self-limited; septicemic/endocarditis forms are progressive and potentially fatal if untreated.

Quality-of-life impact: Erysipeloid causes transient local pain/functional impairment of the hand; endocarditis and musculoskeletal disease (spondylitis/discitis) cause substantial morbidity, disability, and mortality.

4. Genetic / Molecular Information

Human genetics: Not applicable. There are no causal human genes, pathogenic variants (ACMG/AMP), modifier genes, epigenetic lesions, or chromosomal abnormalities — this is an infectious disease, not a heritable disorder.

Pathogen molecular determinants (the relevant "molecular" axis):

- **SpaA (surface protective antigen A)** — a cell-surface adhesin of the **Spa protein family (SpaA, SpaB, SpaC)**. Comparative proteomic/transcriptomic analysis of a virulent strain (HX130709) versus its isogenic avirulent derivative (HX130709a) — profiling 1,299 proteins and 1,673 transcribed genes, with 168 proteins and 475 genes differentially regulated — confirmed **SpaA and neuraminidase, but not hyaluronidase or capsule**, are associated with virulence (F002; [PMID: 27479071](#)).

- **Neuraminidase (sialidase)** — an enzyme acting as a **pathogenicity/spreading factor**, confirmed by rabbit skin test; substrate affinity increases with glycoprotein molecular weight (F002; [PMID: 16869504](#)).
- **Spa family classification:** SpaA, SpaB, SpaC, with **SpaC the most broadly cross-protective antigen** (F006; [PMID: 20926696](#)).

Epigenetics / chromosomal abnormalities: Not applicable.

5. Environmental Information

- **Environmental reservoir/persistence:** The organism persists long-term in the environment, including soil and marine locations, and is a commensal/pathogen across many animal species (F005; [PMID: 19733019](#)).
 - **Occupational/lifestyle factors:** animal handling, fishing/fish processing, butchery, veterinary work; chronic alcohol use is both a lifestyle factor and an immunocompromising comorbidity contributing to invasive disease ([PMID: 24772603](#)).
 - **Infectious agent (taxonomy):** *Erysipelothrix rhusiopathiae* (NCBI txid1648), family *Erysipelotrichaceae*. A related species, *E. piscisicarius*, is relevant to fish and to identification challenges ([PMID: 42670169](#)).
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6. Mechanism / Pathophysiology

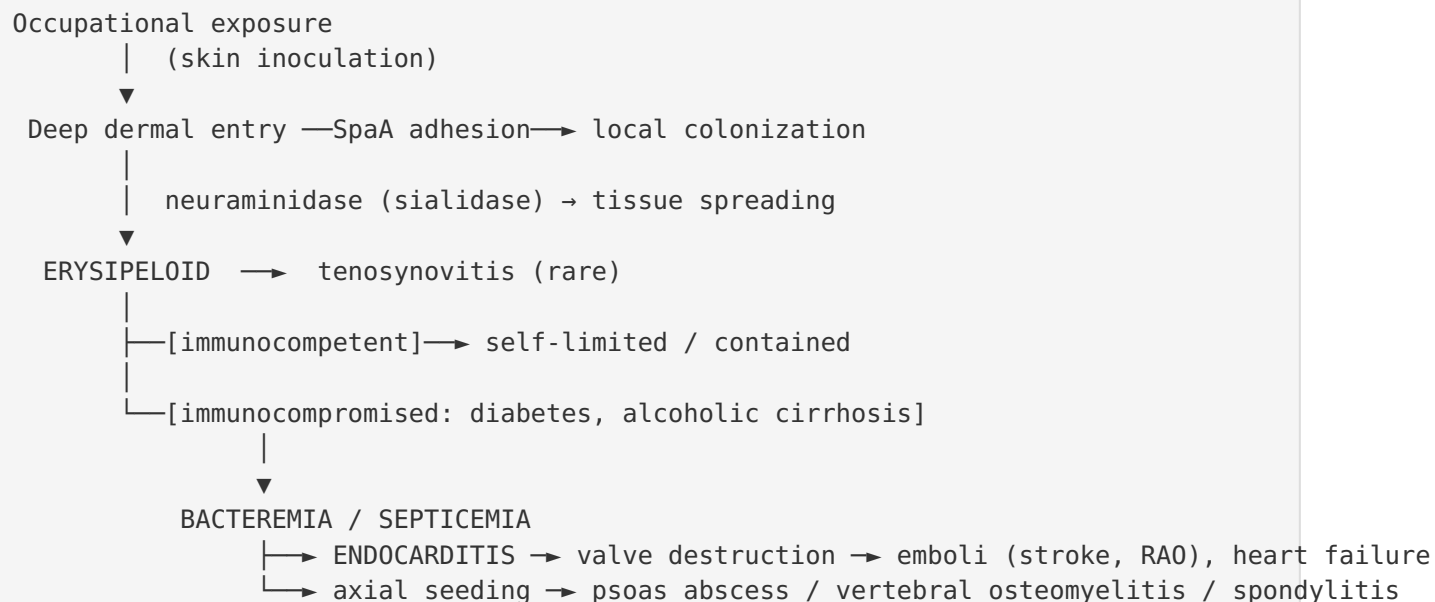
Ordered causal chain (initiating exposure → clinical manifestation)

1. **Occupational contact** with a colonized/infected animal, its products/wastes, or contaminated soil/water **leads to** deposition of *E. rhusiopathiae* on the skin (F001, F008).
2. **Direct inoculation through a cut/abrasion** (usually hand/finger) **results in** entry of the bacterium into the deep dermis, where it resides (F008; [PMID: 26693283](#), [PMID: 10482289](#)).
3. **SpaA-mediated adhesion** to host tissue **promotes** local colonization (F002, F006) — *inferred* from virulence-factor association plus vaccine-protection data.
4. **Neuraminidase (sialidase) cleaves sialic acid** from host glycoproteins, **facilitating** tissue spreading and local invasion (F002; confirmed by rabbit skin test, [PMID: 16869504](#)).

5. Local spread **produces** the characteristic **erysipeloid** plaque; peripheral extension can **lead to tenosynovitis** of the digit (F008; [PMID: 26693283](#)).

↳ **Branch A (contained):** In immunocompetent hosts the infection remains localized and is typically self-limited.

↳ **Branch B (dissemination):** In immunocompromised hosts (diabetes, alcoholic cirrhosis), local containment fails. 6. **Host immunocompromise permits** breach into the bloodstream → **bacteremia/septicemia** (F007; [PMID: 24772603](#)). 7. Circulating bacteria **seed heart valves** (often native, previously normal) → **infective endocarditis** with vegetations (F004). 8. Valve colonization **causes extensive valve destruction** and **generates septic emboli** → stroke, retinal artery occlusion, and heart failure (F004; [PMID: 40619580](#), [PMID: 40087746](#)). 9. Alternatively, hematogenous seeding of the axial skeleton → **psoas abscess, vertebral osteomyelitis, discitis, pyogenic spondylitis** (F007; [PMID: 24772603](#), [PMID: 41111798](#)). 10. Despite low intrinsic virulence, endocarditis **results in** high case-fatality (>38%); low virulence explains the *rarity* of intracranial (embolic) manifestations relative to more aggressive organisms (F004; [PMID: 40619580](#)).



Upstream vs downstream: Adhesion (SpaA) and enzymatic spreading (neuraminidase) are **upstream**; septicemia, valve destruction, and embolic organ injury are **downstream**. Host immune status is the pivotal **gate** determining which branch is taken.

Cell types / processes involved: keratinocytes/dermal fibroblasts and dermis (erysipeloid); vascular endothelium and cardiac valve endothelium (endocarditis); neutrophil-driven inflammation and abscess formation (musculoskeletal disease). Suggested GO terms: GO:0007155 (cell adhesion), GO:0006954 (inflammatory response), GO:0052126 (movement in host — spreading), GO:0016997 (exo- α -sialidase activity), GO:0044409 (entry into host). Suggested CL terms: CL:0000312 (keratinocyte), CL:0000115 (endothelial cell), CL:0000775 (neutrophil).

Molecular pathways / metabolic changes / omics: No canonical human signaling pathway (Wnt/MAPK/mTOR) is implicated. The relevant molecular biology is bacterial (adhesin display, sialidase activity). The comparative proteomic/transcriptomic study (F002) is the key omics dataset, revealing a differentially regulated virulence repertoire.

7. Anatomical Structures Affected

Organ level: - **Primary:** Skin — especially hands/fingers (erysipeloid). UBERON:0002097 (skin of body); UBERON:0002389 (manual digit). - **Secondary (invasive disease):** Heart valves (endocarditis; UBERON:0000946 cardiac valve, UBERON:0002135 mitral valve, UBERON:0002137 aortic valve); vertebral column / intervertebral disc (UBERON:0002228 vertebra; UBERON:0002103 intervertebral disc); psoas muscle (UBERON:0001096); brain vasculature and retina via emboli (UBERON:0000178 blood; UBERON:0000966 retina). - **Body systems:** integumentary, cardiovascular, musculoskeletal, and (secondarily, via emboli) nervous and visual systems.

Tissue/cell level: epithelial/dermal tissue (skin), endothelial tissue (valves, vessels), connective/bone tissue (spondylitis). CL:0000312 (keratinocyte), CL:0000115 (endothelial cell), CL:0000062 (osteoblast, in osteomyelitis).

Subcellular level: Host cell-surface sialylated glycoproteins are the substrate of bacterial neuraminidase. GO cellular component: GO:0009986 (cell surface); GO:0005886 (plasma membrane).

Lateralization: Erysipeloid is typically **unilateral** (site of inoculation, usually dominant hand); endocarditis/embolic events may be **bilateral** (e.g., bilateral retinal artery occlusion reported — [PMID: 40087746](#)).

8. Temporal Development

- **Onset:** Adult-onset (occupational); **acute** for erysipeloid (days after inoculation); **subacute** for endocarditis and musculoskeletal disease.
 - **Progression / course:** Erysipeloid is usually **self-limited** (resolves over weeks, faster with antibiotics). The septicemic/endocarditis form is **progressive** and, untreated, potentially fatal; it frequently causes valve destruction (F004). Musculoskeletal disease (spondylitis/discitis) follows a **subacute-to-chronic** course.
 - **Remission:** Erysipeloid may remit spontaneously; invasive forms require treatment-induced remission (antibiotics ± surgery).
 - **Critical periods / windows of intervention:** Early recognition is crucial — because the organism is **intrinsically vancomycin-resistant**, empiric vancomycin fails, so timely correct identification and penicillin therapy is the key intervention window (F003; [PMID: 12408347](#)).
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9. Inheritance and Population

Inheritance: Not applicable — infectious, non-heritable. No penetrance, expressivity, anticipation, mosaicism, founder effects, consanguinity, or carrier frequency considerations apply.

Epidemiology: Human invasive disease is **rare**; endocarditis and systemic infection are reported chiefly as case reports/series. Precise population incidence/prevalence figures are not well established because the disease is uncommon and under-reported; erysipeloid is more frequent but often self-treated and undocumented.

Population demographics: - **Occupational groups** (butchers, fishermen, fish handlers, veterinarians, farmers, abattoir/poultry workers) are the affected population — an occupational rather than ethnic distribution. - **Sex ratio:** Male predominance is expected given occupational exposure patterns (not precisely quantified in the reviewed evidence). - **Age:** Adults of working age. - **Geographic distribution:** Worldwide, wherever animal husbandry, fishing, and meat processing occur; the organism persists in soil and marine environments globally (F005).

10. Diagnostics

Microbiological identification (diagnostic signature — F009): Clinical isolates are **Gram-positive pleomorphic rods, catalase-negative and oxidase-negative**, PYR-positive, LAP-positive, alpha-hemolytic, and characteristically **produce H₂S in triple sugar iron (TSI) agar** (PMID: 16178460). **Intrinsic vancomycin resistance** is a practical identification clue (one of few vancomycin-resistant Gram-positive rods) that simultaneously mandates a therapy change.

Test	Result for <i>E. rhusiopathiae</i>	Diagnostic value
Gram stain	Gram-positive slender/pleomorphic rod	Screening
Catalase	Negative	Distinguishes from <i>Corynebacterium</i>
Oxidase	Negative	Supportive
H ₂ S in TSI	Positive (characteristic)	Key differentiator (PMID: 16178460)
Hemolysis (blood agar)	Alpha (greening)	Supportive
Vancomycin	Resistant (MIC90 64 mg/L)	Clue + therapy trigger (PMID: 12408347)

Molecular diagnostics: 16S rDNA sequencing lacks resolution to distinguish among *Erysipelothrix* species; long-read (Oxford Nanopore) direct sequencing of clinical tissue can accurately identify species where MALDI-TOF and 16S misassign them (F009; PMID: 42670169).

Imaging / functional tests (invasive disease): Echocardiography (transthoracic/transesophageal) for vegetations and valve destruction in suspected endocarditis; MRI/CT for spondylitis, discitis, and psoas abscess; fundoscopy for embolic retinal artery occlusion.

Blood cultures are the mainstay for septicemic/endocarditis diagnosis. Erysipeloid diagnosis is often **clinical**, because culture is slow and the organism resides deep in the skin, contributing to under-diagnosis (PMID: 10482289).

Differential diagnosis: cellulitis/erysipelas (streptococcal), other bacterial hand infections, **erysipeloid cutaneous leishmaniasis** (an unrelated dermatologic mimic sharing the "erysipeloid" descriptor — PMID: 35609142), and other Gram-positive-rod bacteremias (*Listeria*, *Corynebacterium*, *Lactobacillus*).

Genetic testing / newborn screening / omics diagnostics: Not applicable to a non-genetic infection (beyond pathogen sequencing above).

11. Outcome / Prognosis

- **Erysipeloid:** Excellent prognosis; usually self-limited and fully recovers, faster with penicillin.
- **Endocarditis / septicemic form: Poor** — mortality can **exceed 38% even with appropriate treatment**, with frequent extensive valve destruction (F004; PMID: 40619580). Because it often affects native/previously normal valves, valve replacement may be required.
- **Embolic complications:** stroke and (bilateral) retinal artery occlusion are recognized (PMID: 40619580, PMID: 40087746); the organism's **low virulence** paradoxically makes intracranial embolic manifestations *uncommon* relative to more aggressive pathogens (F004).
- **Musculoskeletal disease:** Recoverable with prolonged antibiotics ± drainage, but under-recognized and can cause lasting spinal morbidity (PMID: 24772603, PMID: 41111798).

Prognostic factors: host immune status (diabetes, cirrhosis, alcohol use worsen outcomes), timeliness of correct diagnosis (avoiding vancomycin failure), valvular involvement, and embolic events.

12. Treatment

Pharmacotherapy (first-line): Penicillin is the drug of choice; ceftriaxone is an effective alternative. Agar-dilution testing of 60 isolates showed penicillin and ceftriaxone highly active (MIC₉₀ 0.03 and 0.125 mg/L) and ciprofloxacin very active (MIC₉₀ 0.06 mg/L) (F003; PMID: 12408347).

Critical caveat — intrinsic vancomycin resistance: The organism is **resistant to vancomycin (MIC90 64 mg/L)** and to gentamicin in reported cases — so standard empiric therapy for Gram-positive bacteremia/endocarditis (vancomycin ± aminoglycoside) will fail (F003; [PMID: 12408347](#), [PMID: 16178460](#)). Early species identification is therefore therapeutically decisive.

Agent	Activity	Role	NCIT suggestion
Penicillin G	MIC90 0.03 mg/L	First-line	NCIT:C61785 (Penicillin)
Ceftriaxone	MIC90 0.125 mg/L	Alternative / severe disease	NCIT:C47616 (Ceftriaxone)
Ciprofloxacin	MIC90 0.06 mg/L	Alternative	NCIT:C376 (Ciprofloxacin)
Vancomycin	MIC90 64 mg/L (resistant)	Avoid	NCIT:C1281 (Vancomycin)

Surgical/interventional: Valve replacement for destructive endocarditis; drainage of psoas abscess and surgical management of vertebral osteomyelitis/discitis when indicated.

Supportive care: wound care for erysipeloid; standard sepsis/endocarditis supportive management for invasive disease.

Pharmacogenomics / gene / cell / RNA / immunotherapy: Not applicable to this bacterial infection.

13. Prevention

Primary prevention (human): Occupational hygiene — protective gloves, prompt cleaning/care of cuts and abrasions, and safe handling of animals/fish/meat. No licensed human vaccine exists.

Primary prevention (animal reservoir — One Health): Spa-family vaccines are the cornerstone of controlling the swine reservoir, indirectly reducing human exposure. Spa proteins comprise SpaA/SpaB/SpaC, with **SpaC the most broadly cross-protective**; mice immunized with rSpaC664 or its N-terminal α -helical domain (rSpaC427) — but not the C-terminal domain — were protected against challenge with serovars 1a, 2, 6, 19, and 18, and

pigs immunized with SpaC427 were protected against the highly virulent heterologous strain Fujisawa (serovar 1a) (F006; [PMID: 20926696](#)). Passive transfer of anti-rSpaC427 serum protected mice (antibody-mediated protection). A recombinant **DnaK+SpaA** formulation also conferred protection and delayed symptoms in a swine-erysipelas model (F006; [PMID: 37815427](#)).

Secondary/tertiary prevention: Early recognition of erysipeloid and prompt penicillin to prevent progression; timely correct microbiological identification to avoid vancomycin failure in invasive disease; endocarditis prophylaxis considerations in high-risk hosts.

Public health / environmental interventions: Herd vaccination, animal husbandry hygiene, carcass/waste management, and worker education in high-risk industries.

14. Other Species / Natural Disease

E. rhusiopathiae is a genuine **broad-host-range zoonotic pathogen** and the veterinary burden dwarfs the human one.

- **Taxonomy of affected hosts:** swine (*Sus scrofa*), turkeys, chickens, ducks, emus, sheep/lambs, plus many wild and domestic mammals, birds, and fish; the bacterium is a "pathogen or commensal in a wide variety of wild and domestic animals, birds and fish" (F005; [PMID: 19733019](#)).
- **Principal animal disease: Swine erysipelas** — the disease "of greatest prevalence and economic importance" (F005; [PMID: 19733019](#)). Manifestations include acute septicemia, characteristic "diamond skin" urticarial lesions, chronic arthritis, and endocarditis in pigs. Other diseases: **erysipelas of farmed turkeys, chickens, ducks, and emus**, and **polyarthritis in sheep and lambs** ([PMID: 10482289](#)).
- **Comparative pathology:** Human endocarditis and animal (porcine) endocarditis/arthritis share virulence mechanisms (Spa adhesins, neuraminidase), enabling shared vaccine antigens across species.
- **Zoonotic potential:** High — human disease is a direct zoonosis from animal contact; related species *E. piscicarius* affects fish and complicates identification ([PMID: 42670169](#)).

15. Model Organisms

- **Mouse model:** The workhorse for vaccine/immunogenicity studies. Active immunization with rSpaC664/rSpaC427 protected mice against multi-serovar challenge; passive serum transfer confirmed antibody-mediated protection (F006; [PMID: 20926696](#)).
- **Pig (natural host) challenge model:** Pigs immunized with SpaC427 were protected against the highly virulent heterologous strain **Fujisawa** — a model that faithfully recapitulates the natural disease and is the gold standard for swine-erysipelas vaccine evaluation (F006; [PMID: 20926696](#), [PMID: 37815427](#)).
- **Rabbit skin test:** Used to demonstrate the spreading/pathogenicity function of neuraminidase in vivo (F002; [PMID: 16869504](#)).
- **Isogenic bacterial strain pair:** Virulent HX130709 vs avirulent HX130709a enabled comparative omics that pinpointed SpaA/neuraminidase as virulence factors (F002; [PMID: 27479071](#)).

Phenotype recapitulation: Mouse and especially pig challenge models reproduce protective immunity and lethal challenge well. **Limitation:** models focus on vaccine protection and swine disease; the sporadic, host-status-gated *human* invasive endocarditis is not modeled directly, and no dedicated model captures the human immunocompromised-host dissemination pathway.

Key Findings (with statistical evidence)

F001 — Three clinical forms; erysipeloid is most common

Two authoritative reviews independently enumerate three recognized human disease forms: localized cutaneous **erysipeloid** (most common), a **diffuse/generalized cutaneous** form, and a **septicemic** form frequently associated with **endocarditis**. Human infection is occupationally acquired via animal/product/waste/soil contact. *"Three forms of human disease have been recognised... a localised cutaneous lesion form, erysipeloid, a generalised cutaneous form and a septicemic form often associated with endocarditis"* ([PMID: 19733019](#)); corroborated by [PMID: 10482289](#).

F002 — SpaA and neuraminidase are the key virulence factors (not capsule or hyaluronidase)

Comparative proteomic/transcriptomic analysis of virulent HX130709 vs isogenic avirulent HX130709a (1,299 proteins; 1,673 transcribed genes; 168 proteins and 475 genes differentially regulated) confirmed **SpaA and neuraminidase — but not hyaluronidase or capsule — are associated with virulence**: *"Our study confirms that SpaA and neuraminidase, but not hyaluronidase and capsule, are associated with virulence in E. rhusiopathiae"* (PMID: 27479071). Neuraminidase's spreading role was confirmed by rabbit skin test (PMID: 16869504).

F003 — Intrinsic vancomycin resistance; penicillin is first-line

Agar-dilution testing of 60 isolates: penicillin/ceftriaxone highly active (MIC₉₀ 0.03 / 0.125 mg/L), ciprofloxacin very active (MIC₉₀ 0.06 mg/L), **vancomycin resistant (MIC₉₀ 64 mg/L)**. *"Erysipelothrix rhusiopathiae is still resistant to vancomycin (MIC₉₀ 64 mg/l)"* and *"Penicillin and ceftriaxone... remained active"* (PMID: 12408347).

F004 — Endocarditis: low-virulence organism, high mortality (~38%)

"Even with appropriate treatment, mortality from endocarditis associated with E. rhusiopathiae can exceed 38%" and *"The low virulence of the E. rhusiopathiae pathogen explains the minimal frequency of intracranial manifestations"* (PMID: 40619580). Frequent valve destruction; embolic events reported.

F005 — Broad host range; swine erysipelas is the most economically important animal disease

"It is a pathogen or a commensal in a wide variety of wild and domestic animals, birds and fish. Swine erysipelas... is the disease of greatest prevalence and economic importance" (PMID: 19733019); also affects turkeys, chickens, ducks, emus, sheep (PMID: 10482289).

F006 — Spa family (SpaA/B/C) underpins vaccines; mouse and pig models validate protection

"...Spa proteins... can be classified into three molecular species — SpaA, SpaB, and SpaC — and... SpaC is the most broadly cross-protective"; "Pigs immunized with SpaC427... were protected from challenge with the highly virulent heterologous E. rhusiopathiae strain Fujisawa (serovar 1a)" (PMID: 20926696). DnaK+SpaA also protective (PMID: 37815427).

F007 — Host immunocompromise predisposes to invasive disease

Invasive bacteremia with psoas abscess, vertebral osteomyelitis and discitis (L2–L3) in a patient with diabetes mellitus and alcoholic cirrhosis: *"He had underlying diabetes mellitus and alcoholic cirrhosis"* and *"Erysipelothrix infection should be considered as a causative pathogen of musculoskeletal infection in immunocompromised patients"* (PMID: 24772603).

F008 — Erysipeloid is a localized, direct-inoculation hand/finger infection that can extend to tenosynovitis

"Erysipelothrix rhusiopathiae is a Gram-positive bacterium that in humans causes skin infections, such as erysipeloid, as a result of direct contact with contaminated animals or their waste or products" (PMID: 26693283). Deep-skin residence complicates culture (PMID: 10482289).

F009 — Diagnostic signature: catalase/oxidase-negative Gram-positive rod, H₂S-positive in TSI, vancomycin-resistant

"...the production of H₂S in triple sugar iron agar, was demonstrated" (PMID: 16178460); *"16S rDNA sequencing lacks sufficient resolution to differentiate among Erysipelothrix species"* (PMID: 42670169).

Mechanistic Model / Interpretation

The disease is best understood as a **two-stage, host-gated process**. Stage 1 is **local**: after occupational inoculation, the bacterium exploits **SpaA-mediated adhesion** and **neuraminidase-mediated sialic-acid cleavage/spreading** to establish an erysipeloid plaque in the deep dermis. This stage is typically self-limiting in immunocompetent hosts. Stage 2 is **invasive** and occurs primarily when host defenses are compromised (diabetes, alcoholic cirrhosis): the same virulence machinery permits **bloodstream invasion**, followed by **valve seeding (endocarditis)** or **axial skeletal seeding (spondylitis/discitis/psoas abscess)**. The paradox of a "low-virulence" organism causing >38% endocarditis mortality is resolved by noting that low virulence limits *embolic aggressiveness* (fewer intracranial events) while chronic valve colonization still produces lethal *structural* valve destruction. Clinically, the single most consequential mechanistic fact is **intrinsic vancomycin resistance**: the standard

empiric Gram-positive regimen is ineffective, so the diagnostic signature (catalase/oxidase-negative rod, H₂S-positive TSI, vancomycin-resistant) must be recognized quickly to switch to penicillin.

Evidence Base

PMID	Type	Supports
19733019	Review	Three clinical forms; broad host range; swine erysipelas (F001, F005)
10482289	Review	Three forms; affected animals; deep-skin diagnostic difficulty (F001, F005, F008)
27479071	Comparative omics (isogenic strains)	SpaA/neuraminidase = virulence; capsule/hyaluronidase not (F002)
16869504	Enzymology + rabbit skin test	Neuraminidase as spreading/pathogenicity factor (F002)
12408347	Susceptibility study (60 isolates)	Vancomycin resistance; penicillin/ceftriaxone first-line (F003)
40619580	Case report/review	Endocarditis mortality >38%; low-virulence paradox (F004)
20926696	Vaccinology (mouse + pig)	Spa family; SpaC cross-protection; pig challenge (F006)
37815427	Vaccinology (swine model)	DnaK+SpaA protection (F006)
24772603	Case report	Immunocompromise → invasive musculoskeletal disease (F007)
26693283	Case report/review	Erysipeloid = direct inoculation; tenosynovitis (F008)
16178460	Case report	H ₂ S-in-TSI; vancomycin/gentamicin resistance (F009, F003)
42670169	Genomics (Nanopore)	16S limitation; long-read identification (F009)
40087746	Case report	Bilateral retinal artery occlusion / embolic endocarditis
41111798	Case report	Pyogenic spondylitis (zoonotic)

Challenging / confounding literature: Several retrieved papers on "*erysipeloid cutaneous leishmaniasis*" (PMID: [35609142](#), PMID: [33236713](#)) describe an unrelated *Leishmania* dermatologic entity that merely shares the "erysipeloid" descriptor — a useful reminder for differential diagnosis but not evidence about *E. rhusiopathiae*.

Limitations and Knowledge Gaps

- **Rare-disease evidence base:** Human invasive disease evidence is dominated by case reports; there are **no controlled human incidence/prevalence or mortality datasets**, so epidemiologic figures (except the ~38% endocarditis case-fatality) and sex/age ratios are imprecise.
- **Not a genetic disease:** Sections on causal genes, pathogenic variants, inheritance, penetrance, epigenetics, and newborn/genetic screening are **not applicable**; the "molecular" content is pathogen biology.
- **Host-susceptibility mechanism unquantified:** The dissemination "gate" is described qualitatively (diabetes, cirrhosis, alcohol) without a quantified relative risk or mechanistic immunology.
- **SpaA adhesion step is inferred:** SpaA's causal role rests on virulence-association and vaccine-protection data rather than a direct receptor-binding demonstration in human tissue.
- **No human vaccine:** Vaccine evidence is entirely veterinary (mouse/pig); translation to humans is untested and likely unnecessary given rarity.
- **Identification pitfalls:** MALDI-TOF and 16S can misassign *Erysipelothrix* species; species-level epidemiology (*E. rhusiopathiae* vs *E. piscicarius*) may be under-resolved in older literature.

Proposed Follow-up Experiments / Actions

1. **Quantify host-susceptibility risk:** Systematic review/meta-analysis of invasive-disease case reports to estimate the association between diabetes/alcoholic cirrhosis and dissemination, and to refine the endocarditis case-fatality estimate.
2. **Define SpaA/neuraminidase mechanism directly:** Identify the human host receptor(s) for SpaA and map neuraminidase substrate specificity on cardiac valve endothelium; test isogenic *spaA* and *nanH* knockouts in an endocarditis model.
3. **Improve species-level diagnostics:** Prospectively benchmark long-read (Nanopore) direct-from-tissue sequencing vs MALDI-TOF/16S for *Erysipelothrix* species assignment ([PMID:](#)

[42670169](#)) and disseminate a rapid "vancomycin-resistant Gram-positive rod → suspect *Erysipelothrix* → start penicillin" clinical alert.

4. **Optimize One-Health vaccines:** Extend SpaC-based cross-serovar protection studies ([PMID: 20926696](#)) and DnaK+SpaA formulations ([PMID: 37815427](#)) to broader field serovars to reduce the swine reservoir and, indirectly, human exposure.
5. **Occupational surveillance:** Establish targeted reporting in high-risk industries (abattoirs, fisheries, veterinary) to generate the missing incidence data.

Report compiled from 9 confirmed findings across 23 reviewed papers over 5 investigation iterations. Evidence types span authoritative reviews, comparative bacterial omics, susceptibility testing, veterinary vaccinology (mouse and pig models), and human clinical case reports.

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